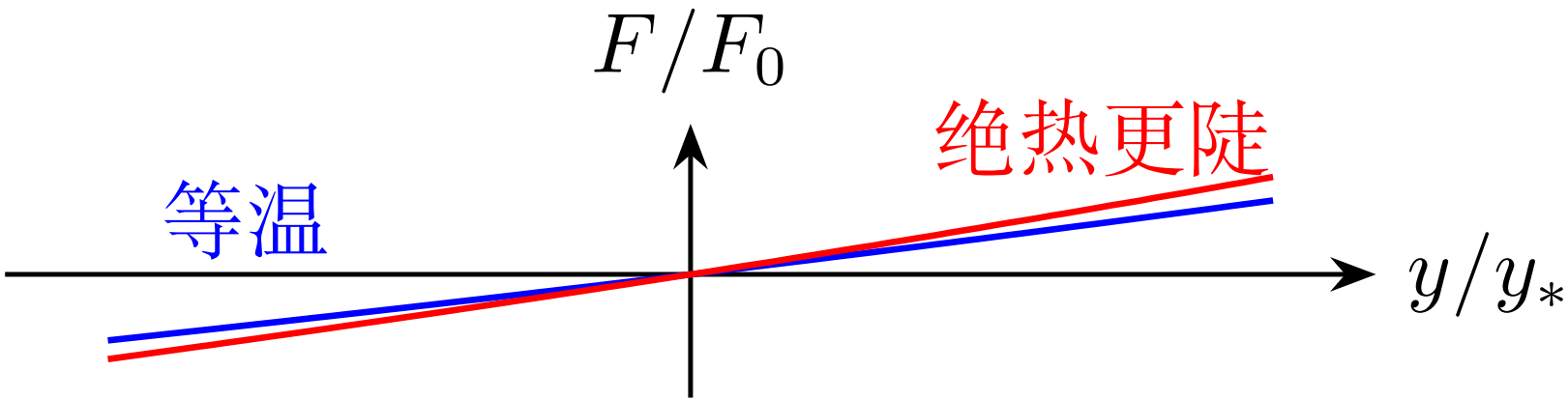

$$dF/dy = A_e\, dp/dy + \beta(p - p_0) > 0$$